

1 (2.11) verification

The admissible weight property

$$(G(t, \cdot) * \rho(\cdot))(x) = (2\pi t)^{-d/2} \int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) \rho(y) dy \leq K_T \rho(x) \quad \forall x \in \mathbb{R}^d, K_T \in \mathbb{N}, t \in [0, T]$$

tells us that

$$\lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq K_T \quad t \in [0, T]. \quad (1.1) \quad \boxed{\text{aw}}$$

The expression for K_t should be

$$K_t := \sup_{x \in \mathbb{R}^d} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)}$$

The limit in (1.1) gives a sufficient (also necessary) condition for K_t to be finite, but in general, this limit is not equal to the value of K_t .

I meant for K_T to mean $C_\rho(T)$. I made some corrections above

We want to show that

$$\exp(-a|x|^2) \leq K'_a \rho(x) \quad (1.2) \quad \boxed{\text{con}}$$

Why does (1.1) imply (1.2)?

(1.4) and (1.1) imply (1.3) which implies (1.2)

and it suffices to show that

$$\lim_{|x| \rightarrow \infty} \frac{\exp(-a|x|^2)}{\rho(x)} \leq K'_a. \quad (1.3) \quad \boxed{\text{con1}}$$

since then

$$\exp(-a|x|^2) \leq (K'_a + \epsilon) \rho(x) \quad |x| > R_\epsilon$$

and by the positivity and continuity of both $\exp(-a|x|^2)$ and $\rho(x)$, we know that there exists some positive constant $K(a, \epsilon)$ such that

$$\exp(-a|x|^2) \leq K'(a, \epsilon) \rho(x) \quad x \leq R_\epsilon$$

and by then letting $K(a, \epsilon) = \max\{K'(a, \epsilon), (K'_a + \epsilon)\}$ then

$$\exp(-a|x|^2) \leq K(a, \epsilon) \rho(x) \quad x \in \mathbb{R}^d$$

which is what we want to show.

In order to show (1.3), it suffices to show that

$$\lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\exp(-a|x|^2)} = \infty \quad (1.4) \quad \boxed{\text{con2}}$$

because this would mean that $\exp(-a|x|^2)$ is much smaller than $(G(t, \cdot) * \rho(\cdot))(x)$ for all large $|x|$ and since (1.1) is true then this would mean that (1.3) would be true as well by squeeze theorem,

$$0 \leq \lim_{|x| \rightarrow \infty} \frac{\exp(-a|x|^2)}{\rho(x)} \leq \lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq K_t$$

(1.4) is proved as follows: first recall that $|\rho(x)| < M$ for some $M \in \mathbb{N}$. Then,

$$0 \leq \frac{\int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) \rho(y) dy}{\exp(-a|x|^2)} \leq \frac{M \int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) dy}{\exp(-a|x|^2)}$$

$$\begin{aligned}
&= \frac{M * \sqrt{2\pi t}}{\exp(-a|x|^2)} \\
&\rightarrow \infty \quad \text{as } |x| \rightarrow \infty
\end{aligned}$$

Thus (1.4) is proved which means that (1.2) is also true.

Can you give an explicit form of $C_\rho(T)$ so that we can check whether $\sup_{T>0} C_\rho(T) < \infty$, or your K_t ? If not, can you derive some sufficient conditions that guarantee this property?

In the above wrote K_t to mean the same thing as $C_\rho(T)$. However in general there is no explicit form for this constant and also in general, $\sup_{T \geq 0} K_T \not\leq \infty$. For example, when we consider the weight $\rho(x) = \exp(-|x|)$ in $\mathbb{R}$ then we can show that $(\bar{G}(t, \cdot) * \exp(-|\cdot|))(x) \leq 2 \exp(t/2) \exp(-|x|)$ and clearly $\sup_{t \geq 0} 2 \exp(t/2) = \infty$.

This is a good example. We should not expect $\sup_{T>0} C_\rho(T) < \infty$. Actually, you may want to check eq. (4.4) of my AOP paper with Robert 2015 on p. 3040.

2 Mistake involving the $G(t, x) \leq KG(1, x)$ inequality

During a meeting we had, we proved that $G(t, x) \leq KG(1, x)$ where the constant K does not depend on x or t but we made a mistake in doing so. What we did is the following (for now we work in $\mathbb{R}$):

$$\begin{aligned}
\frac{G(t+1, x)}{G(1, x)} &= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2(t+1)} + \frac{|x|^2}{2}\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \left(-\frac{1}{t+1} + 1\right)\right) \quad \text{mistake was made here when factoring} \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \frac{t}{t+1}\right) \\
&\leq K
\end{aligned}$$

However if we fix the mistake, what we should get is that

$$\begin{aligned}
\frac{G(t+1, x)}{G(1, x)} &= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2(t+1)} + \frac{|x|^2}{2}\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \left(\frac{1}{t+1} - 1\right)\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(\frac{|x|^2}{2} \frac{t}{t+1}\right) \\
&\not\leq K
\end{aligned}$$

Let's remove the property $\sup_{t \geq 0} \mathbb{E} \left(|u(t, \cdot)|_\rho^2 \right) < \infty$ unless you can show that $C_\rho(T)$ is bounded in T .

3 Mistake that $G(t + t_0, x)$ is bounded above by $G(t, x)$ for $t > t_0$

$$\begin{aligned}
\frac{G(t + t_0, x)}{G(t_0, x)} &= \sqrt{\frac{t}{t + t_0}} \exp\left(-\frac{|x|^2}{2(t + t_0)}\right) \exp\left(\frac{|x|^2}{2t_0}\right) \\
&= \sqrt{\frac{t_0}{t + t_0}} \exp\left(\frac{|x|^2}{2(t + t_0)}\right)
\end{aligned}$$

and this is not bounded above by a constant independent of x and t .

This makes more sense!

4 Admissible Weight

4.1 $\rho(x) = \exp(-|x|)$

Let $\rho(x) = \exp(-|x|)$.

$$\begin{aligned}
 \sqrt{2\pi t} (G(t, \cdot) * \rho)(x) &= \int_{\mathbb{R}} \exp\left(-\frac{|y|^2}{2t}\right) \exp(-|x-y|) dy \\
 &\leq \int_{\mathbb{R}} \exp\left(-\frac{|y|^2}{2t}\right) \exp(-|x| + |y|) dy \quad \text{since } -|x-y| \leq -|x| + |y| \\
 &= \exp(-|x|) \int_{\mathbb{R}} \exp\left(-\frac{|y|^2}{2t}\right) \exp(|y|) dy \\
 &= \exp(-|x|) \int_{\mathbb{R}} \exp\left(-\frac{|y|^2 - 2t|y|}{2t}\right) dy \\
 &= \exp(-|x|) \exp(t/2) \int_{\mathbb{R}} \exp\left(-\frac{(|y| - t)^2}{2t}\right) dy \\
 &= \exp(-|x|) \exp(t/2) \sqrt{2\pi t} \left[\operatorname{erf}\left(\sqrt{\frac{t}{2}}\right) + 1 \right]
 \end{aligned}$$

and therefore

$$(G(t, \cdot) * \rho)(x) \leq \rho(x) \exp(t/2) \left[\operatorname{erf}\left(\sqrt{\frac{t}{2}}\right) + 1 \right].$$

Now fix $T > 0$. Since $\exp(t/2) \left[\operatorname{erf}\left(\sqrt{t/2}\right) + 1 \right]$ is increasing in t , then for every $t \in [0, T]$ we have that

$$(G(t, \cdot) * \rho)(x) \leq \exp(T/2) \left[\operatorname{erf}\left(\sqrt{\frac{T}{2}}\right) + 1 \right] \rho(x)$$

which shows that $\rho(x) = \exp(-|x|)$ is admissible with $C_\rho(T) = \exp(T/2) \left[\operatorname{erf}\left(\sqrt{\frac{T}{2}}\right) + 1 \right]$, which is the sharpest such constant for the particular ρ given in this example.

4.2 $\rho(x) = \exp(-|x|^2)$

In this case

$$\rho(x) = \sqrt{\pi} G(1/2, x)$$

and so we see that

$$(G(t, \cdot) * \rho)(x) = \sqrt{\pi} (G(t, \cdot) * G(1/2, \cdot))(x) = \sqrt{\pi} G(t + 1/2, x) = \frac{1}{\sqrt{2t+1}} \exp\left(\frac{-|x|^2}{2t+1}\right). \quad (4.1)$$

$\mathbf{E} : \exp(x^2)$

I claim that for any $T > 0$, that there does not exist a constant $K(T)$ such that

$$(G(t, \cdot) * \rho)(x) \leq K(T) \exp(-|x|^2) \quad \forall t \in [0, T] \text{ and } x \in \mathbb{R}.$$

BWOC, suppose there did. Then for $T = 4$, there exists some constant $K(4)$ such that for any $t \in [0, 4]$,

$$\frac{1}{\sqrt{2t+1}} \exp\left(\frac{-|x|^2}{2t+1}\right) \leq K(4) \exp(-|x|^2).$$

Note that for any $0 \leq t \leq 4$ and $x \in \mathbb{R}$,

$$\frac{1}{3} \exp\left(\frac{-|x|^2}{2t+1}\right) \leq \frac{1}{\sqrt{2t+1}} \exp\left(\frac{-|x|^2}{2t+1}\right)$$

and so we must have

$$\frac{1}{3} \exp\left(\frac{-|x|^2}{2t+1}\right) \leq K(4) \exp(-|x|^2).$$

This further implies that

$$\begin{aligned} 1 &\leq 3K(4) \exp\left(\frac{|x|^2}{2t+1} - |x|^2\right) \\ &= 3K(4) \exp\left(\frac{|x|^2 - 2t|x|^2 - |x|^2}{2t+1}\right) \\ &= 3K(4) \exp\left(\frac{-2t|x|^2}{2t+1}\right) \end{aligned}$$

and this must hold for all $t \in [0, 4]$ and $x \in \mathbb{R}$. However, as $|x| \rightarrow \infty$ then

$$3K(4) \exp\left(\frac{-2t|x|^2}{2t+1}\right) \rightarrow 0$$

and so

$$1 \leq 3K(4) \exp\left(\frac{-2t|x|^2}{2t+1}\right) \quad (4.2) \quad \boxed{\text{E:contradict.}}$$

cannot hold for all $x \in \mathbb{R}$ which is a contradiction. Thus $\exp(-|x|^2)$ is not admissible.

4.3 $\rho(x) = \exp(-|x|^b)$

Suppose that $b \in (1, 2)$ and that $\rho(x) = \exp(-|x|^b)$.

$$\begin{aligned} (G(t, \cdot) * \rho)(x) &= \int_{\mathbb{R}} G(t, x-y) \exp(-|y|^b) dy \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_{\mathbb{R}} G(t, x-y) |y|^{nb} dy \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \mathbb{E}(|N|^{nb}) \end{aligned}$$

where above N is a normal random variable with mean x and standard deviation t . Note that this can't technically be done because the power series for $\exp(x)$ does not converge uniformly on the whole real line, however doing this seems to work as can be seen in Section 4.4 below. According to [2], this moment is given by

$$\mathbb{E}(|N|^{nb}) = (2t)^{nb/2} \frac{\Gamma\left(\frac{nb+1}{2}\right)}{\sqrt{\pi}} \phi\left(-\frac{nb}{2}, \frac{1}{2}; -\frac{x^2}{2t}\right)$$

where ϕ is the Kummer Confluent Hypergeometric function

$${}_1F_1(\alpha, \gamma; z) = \phi(\alpha, \gamma; z) = \sum_{k=0}^{\infty} \frac{\alpha^{\bar{k}}}{\gamma^{\bar{k}}} \frac{z^k}{k!} \quad \text{and} \quad z^{\bar{k}} = \frac{\Gamma(z+k)}{\Gamma(z)}.$$

Plugging this in above gives us that

$$(G(t, \cdot) * \rho)(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (2t)^{nb/2} \frac{\Gamma\left(\frac{nb+1}{2}\right)}{\sqrt{\pi}} \phi\left(-\frac{nb}{2}, \frac{1}{2}; -\frac{x^2}{2t}\right) \quad (4.3) \quad \boxed{\text{E:exp}(x^{\wedge}b)}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (2t)^{nb/2} \frac{\Gamma\left(\frac{nb+1}{2}\right)}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-nb/2)^{\bar{k}} \left(\frac{-x^2}{2t}\right)^k}{(1/2)^{\bar{k}} k!} \quad (4.4)$$

Note that it seems that (4.3) is only true when $b \in \mathbb{N}$ (see Section (4.5) and equation (4.6)).

If it works for $b \in \mathbb{N}$, it should work for fractions as well. I can show you some numerical evidence that why the case $b > 1$ won't work during our meeting. We may not need to spend too much time here unless we can really prove that ρ with $b > 1$ is not admissible.

4.4 Special case when $x = 0$ and $b = 2$

If we set $b = 2$ and $x = 0$ in (4.3) then we should be able to recover (4.1) with x set to 0, which is

$$\frac{1}{\sqrt{2t+1}}.$$

Letting $b = 2$ and $x = 2$ in (4.3) gives us

$$\sum_{n=0}^{\infty} \frac{(2n)!(-2t)^n}{4^n (n!)^2} \quad (4.5)$$

and plugging this into Wolfram gives us that

$$\sum_{n=0}^{\infty} \frac{(2n)!(-2t)^n}{4^n (n!)^2} = \frac{1}{\sqrt{2t+1}} \quad \text{whenever } |t| \leq 1/2.$$

Another thing to note is that the Borel Regularization of the sum (4.5) is also equal to $1/\sqrt{2t+1}$ for any $t > 0$. This can be easily checked by plugging

`Sum[(-2t)^k (2k)! / (4^k (k!)^2), {k, 0, Infinity}, Regularization -> "Borel"]`

into Mathematica.

Its possible that we may be able to use this technique to calculate (4.3) for more general x and b . In fact, what I believe that we may be able to show is that if we fix any $t > 0$ then $(G(t, \cdot) * |\cdot|^b)(x)$ will decay much faster than $|x|^b$ as $|x| \rightarrow \infty$.

I can show you during our meeting this week that the case $b = 2$ won't work.

4.5 More Mathematica Calculations.

I am unsure that Borel Regularization can be used. For example, I considered $b = 3/2$, $t = 1/2$ and $x = 0$ and if we plug this into (4.3) then this gives us

$$(G(1/2, \cdot) * \rho)(0) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{\Gamma\left(\frac{n(3/2)+1}{2}\right)}{\sqrt{\pi}} \phi\left(-\frac{(3/2)n}{2}, \frac{1}{2}; 0\right)$$

however we get different answers when using Mathematica to calculate this and also this

$$\int_{\mathbb{R}} G(1/2, y) \exp\left(-|y|^{(3/2)}\right) dy.$$

The codes used were

`Sum[((-1)^n Gamma[((3/2)*n+1)/2] Hypergeometric1F1[(-(3/2)*n/2), 1/2, 0]) / (Sqrt[Pi] n!),
{n, 0, Infinity}, Regularization -> "Borel"]`

and

`Integrate[(1/Sqrt[Pi])Exp[-Abs[y]^2]Exp[-Abs[y]^(3/2)],{y,-Infinity,Infinity}]`.

However, it seems that the following is true for every $b \in \mathbb{N}$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{\Gamma\left(\frac{n(b)+1}{2}\right)}{\sqrt{\pi}} \phi\left(-\frac{(b)n}{2}, \frac{1}{2}; 0\right) = \int_{\mathbb{R}} G(1/2, y) \exp(-|y|^b) dy \quad (4.6) \quad \boxed{\text{E:BRsumrep}}$$

where the above sum is the Borel regularization.

Lastly, what I wished to be able to do what calculate

$$\int_{\mathbb{R}} G(t, x - y) \exp(-|y|^b) dy$$

on Mathematica for arbitrary t, x and b but I was not able to do so. The reason it gave me was that it took too much computational time for my account so maybe you could do it with your account. However to simplify this, we could just fix a t value, say $t = 1/2$ and this simplifies the integral to

$$\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \exp(-|x - y|^2) \exp(-|y|^b) dy.$$

However I still cant calculate this on Mathematica. What I believe to be true is that if we assume by contradiction that for $T = 1/2$, there exists some constant $K(1/2)$ such that for any $0 \leq t \leq 1/2$ and $x \in \mathbb{R}$

$$\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \exp(-|x - y|^2) \exp(-|y|^b) dy \leq K(1/2) \exp(-|x|^b)$$

then we will come up with a contradiction. If we use Mathematica, we see that for the case $t = 1/2$ and $b = 2$ then

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \exp(-|x - y|^2) \exp(-|y|^2) dy}{\exp(-|x|^2)} = \infty$$

which again shows that $\exp(-|x|^2)$ is not admissible. But Mathematica would not calculate the corresponding limit for any particular $b \in (1, 2)$. For example, I tried taking $b = 3/2$, which is the limit

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \exp(-|x - y|^2) \exp(-|y|^{3/2}) dy}{\exp(-|x|^{3/2})}$$

and the codes I used were:

`f[x_] := Integrate[(1/Sqrt[Pi])Exp[-Abs[x-y]^2]Exp[-Abs[y]^(3/2)],{y,-Infinity,Infinity}]`

`Limit[f[x]Exp[Abs[x]^(3/2)],x ->Infinity]`

but Mathematica would time out each time I tried.

Here you might want to try “NIntegrate” instead of “Integrate”.

5 Extending the results to the fractional heat equation.

The following equation driven by space-time white noise is studied in [1]

$$\begin{cases} \left(\frac{\partial}{\partial t} - {}_x D_{\delta}^{\alpha}\right) u(t, x) = \rho(u(t, x)) \dot{W}(t, x), & t > 0, \text{ and } x \in \mathbb{R} \\ u(0, \cdot) = \mu(\cdot). \end{cases}$$

The following moment bounds are proven there:

$$\|u(t, x)\|_p^2 \leq \begin{cases} J_0^2(t, x) + ([\bar{\varsigma} + J_0^2] \star \bar{\mathcal{K}})(t, x), & p = 2 \\ 2J_0^2(t, x) + ([\bar{\varsigma} + 2J_0^2] \star \hat{\mathcal{K}}_p)(t, x), & p > 2, \end{cases} \quad (5.1)$$

E:fracheatsp

and $\mathcal{K}$ has the following upper bound

$$\mathcal{K}(t, x) \leq \frac{C}{t^{1/a}} \delta G_a(t, x) \left(1 + t^{1/a} \exp(\gamma^{a^*} t)\right).$$

5.1 Discussion.

1. We run into the issue with this bound not being bounded in t
2. The space time convolution in the moment bound versus just a spacial convolution may cause problems. When dealing with the heat equation and a noise white in time and colored in space, the moment bound only has a spacial convolution.
3. We may be able to study equation (5.1) but with a noise that is white in time and colored in space and get a result similar to the heat equation case and use this to show the existence of an invariant measure.

6 $\Upsilon(\beta)|_{\beta=0} < \infty$ versus the spectral density integrability condition in [TZ98].

Hypothesis 2.1 [TZ98] says that the spectral density $\hat{f}$ must satisfy the property that

$$\hat{f} \in L^p(\mathbb{R}^d) \quad \text{for some } p \in (d/(d-2), \infty] \quad (6.1)$$

E:hyp2.1

Our assumption is that

$$\Upsilon(0) = (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(\xi)}{|\xi|^2} d\xi < \infty, \quad d \geq 3. \quad (6.2)$$

E:speccond

I believe that we may find a $\hat{f}$ that satisfies (6.2) but not (6.1). First, suppose (6.2) is true. Then we know that $\hat{f}(\xi)$ may have some relaxed decay properties at $|\xi| = \infty$ because of the help from the $|\xi|^{-2}$ term. In particular we can have decay as slow as $|\xi|^{-a}$ for $a \in (d-2, \infty)$ at infinity. At the same time, the $|\xi|^{-2}$ forces $\hat{f}$ to have growth no faster than $|\xi|^{-a}$ for $a \in (0, d-2)$ around the origin. So, if we assume $\hat{f}$ satisfies the condition (6.2) then $\hat{f}$ must be integrable around the origin but may not be integrable at infinity because $\hat{f}$ may have tails that decay as slow as $|\xi|^{-a}$ for $a \in (d-2, d)$.

Considering the above, it seems as if we assume $\hat{f}$ satisfies (6.2), then $\hat{f}$ may not be able to satisfy the condition (6.1). First, let $d \geq 3$ and consider $0 < \delta < K$ where K will be determined below. Now consider $\hat{f}$ such that $\hat{f}$ behaves like $|\xi|^{-(d-2-\delta)}$ around the origin and $|\xi|^{-(d-2+\delta)}$ at infinity. BWOC suppose that $\hat{f}$ satisfies condition (6.1). Then to take care of the thick tails, we would need p to be such that

$$p \geq \frac{d}{d-2+\delta}$$

but recall that $p \geq d/(d-2)$ so we must choose a $\gamma > 0$ such that

$$\frac{d+\gamma}{d-2+\delta} > \frac{d}{d-2}$$

and thus we need

$$p \geq \frac{d+\gamma}{d-2+\delta}.$$

However, at the same time we still need to maintain integrability around the origin. But we will now see that if we choose $\delta > 0$ small enough that we will lose integrability around the origin. Note that we may say that

$$p(d-2-\delta) \geq \frac{(d+\gamma)(d-2-\delta)}{d-2}$$

and if we set

$$\frac{(d+\gamma)(d-2-\delta)}{d-2+\delta} \geq d$$

then this is satisfied when

$$\frac{d-2-\delta}{d-2+\delta} \geq \frac{d}{d+\gamma} \tag{6.3}$$

and since $\delta > 0$ is arbitrary and

$$\lim_{\delta \rightarrow 0} \frac{d-2-\delta}{d-2+\delta} = 1 > \frac{d}{d+\gamma}$$

then there exists a $K > 0$, independent of p , such that for any $\delta \in (0, K)$ that (6.3) is true. And therefore this implies that

$$p(d-2-\delta) \geq d$$

and so $\hat{f}$ cannot be integrable around the origin which means that $\hat{f}$ satisfies (6.2) but not (6.1).

References

CDfracheat15

TZ98

Winklebauer

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[TZ98] Tessitore, Gianmario and Jerzy Zabczyk. Invariant measures for stochastic heat equations. *Probab. Math. Statist.* 18 (1998), no. 2, Acta Univ. Wratislav. No. 2111, 271–287.

[2] Winklebauer. Moments and Absolute Moments of the Normal Distribution. *Arxiv* <https://arxiv.org/abs/1209.4340>.